

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Loan Payment Web System

Document Information

Field	Details
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Project Name	Loan Repayment Portal
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Document Purpose

This Business Requirements Document (BRD) defines the business objectives, project scope, stakeholders, business requirements, and high-level functional and non-functional requirements for the Loan Payment Web System. The document serves as the primary reference for stakeholders throughout the project lifecycle.

1. Document Revision History

The following table records the revision history of this Business Requirements Document (BRD). It provides a log of all significant updates made during the project lifecycle.

Version	Date	Author	Description of Changes	Approved By
1.0	06/07/2026	Goutam Kumar	Initial draft of the Business Requirements Document.	
1.1	07/07/2026	Goutam Kumar	Updated scope and requirements to align with implemented release	
Final	08/07/2026	Goutam Kumar	Submission-ready BRD with acceptance mapping and deployment references	

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2. Introduction

The Loan Payment Web System is a web-based application designed to simplify and automate the loan repayment process for borrowers and financial institutions. The system provides a centralized platform where customers can securely view loan details, make online EMI payments, access payment history, and download payment receipts. It also enables administrators to manage customer information, monitor loan accounts, generate reports, and maintain accurate repayment records.

The purpose of this Business Requirements Document (BRD) is to define the business requirements, objectives, scope, stakeholders, and high-level requirements for the Loan Payment Web System. This document serves as a reference for all stakeholders throughout the project lifecycle and ensures that the delivered solution meets the organization's business goals.

2.1 Purpose

The purpose of this Business Requirements Document is to capture and document the business needs for developing the Loan Payment Web System. It establishes a common understanding among business stakeholders, project managers, developers, testers, and end users regarding the expected functionality and objectives of the system.

This document will be used as the primary reference during the analysis, design, development, testing, implementation, and maintenance phases of the project.

2.2 Background

Many financial institutions still rely on manual or semi-automated methods for collecting loan repayments and maintaining repayment records. These traditional processes often result in delayed payment updates, manual errors, limited transparency, and increased operational costs. Customers may also experience inconvenience due to the lack of real-time access to loan information and payment history.

To address these challenges, the Loan Payment Web System is proposed as a secure and user-friendly web application that enables customers to make loan payments online while providing administrators with tools to manage loans, monitor repayments, and generate reports efficiently.

2.3 Business Need

The organization requires a digital solution that streamlines the loan payment process, minimizes manual intervention, improves transaction accuracy, and enhances customer satisfaction. By automating payment processing and maintaining centralized loan records, the organization can improve operational efficiency, reduce administrative effort, and provide better services to its customers.

2.4 Intended Audience

Stakeholder	Purpose of Using this Document
Business Analyst	Defines, analyses, and manages business requirements.
Project Manager	Plans, monitors, and controls project activities.
Development Team	Develops the system based on documented requirements.
Quality Assurance Team	Validates that the implemented solution meets the business requirements.
Client / Business Sponsor	Reviews, approves, and provides business feedback.
System Administrator	Uses the document for system deployment and maintenance planning.

2.5 Implemented Release Snapshot

The current release delivers a customer-facing loan repayment solution with secure identity verification, repayment processing, and transaction visibility.

Implemented Features (Current Release):

- Customer identity verification using registered mobile number and card last 4 digits
- reCAPTCHA validation (with approved demo fallback option for assignment mode)
- Loan detail's view: outstanding amount, minimum due, due date, payment status
- Repayment options: minimum due, total due, and custom amount
- Simulated 1Pay payment flow with OTP verification
- Payment receipt generation after successful transaction
- Payment history view for customer transactions
- Zero-balance handling ("no payment required" state)

Technology Overview:

- Frontend: React + TypeScript + Vite
- Backend: Node.js + Express + TypeScript
- Database: Supabase (PostgreSQL)
- Hosting: Vercel (Frontend), Render (Backend)

3. Business Problem

The current loan repayment process relies on manual or partially automated systems, which often result in delays, data inconsistencies, and limited visibility into customer loan information. Borrowers face challenges in tracking loan balances, payment schedules, and transaction history, while financial institutions spend significant time managing records manually and resolving payment-related issues.

To improve operational efficiency and customer satisfaction, a centralized Loan Payment Web System is required. The proposed solution will automate payment processing, maintain accurate loan records, provide real-time updates, and enable secure online transactions.

3.1 Current Business Challenges

Challenge ID	Business Challenge	Impact
BC-01	Manual loan	Increased processing time and human errors.
BC-02	Lack of centralized loan records	Difficulty tracking customer loan information.
BC-03	Delayed payment updates	Customers cannot view real-time payment status.
BC-04	No automated payment reminders	Increased risk of missed or late EMI payments.
BC-05	Limited reporting capabilities	Management cannot easily monitor repayment performance.
BC-06	Manual receipt generation	Additional administrative workload and delays.

3.2 Existing Process Limitations

- Loan payments are processed manually or through disconnected systems.
- Payment records are updated with delays.
- Customers have limited visibility into outstanding loan balances.
- Administrators spend considerable time verifying and reconciling payments.
- Reports are generated manually, increasing the possibility of errors.
- The existing process lacks timely notifications for due payments.

3.3 Business Impact

The existing loan payment process affects both customers and the organization. Customers may experience delayed payment confirmations, difficulty accessing loan information, and inconvenience in tracking repayment history. The organization faces increased operational costs, inefficient resource utilization, higher administrative effort, and reduced customer satisfaction. Implementing a centralized Loan Payment Web System will address these issues by providing automation, transparency, and real-time information.

4. Business Objectives

The objective of this project is to develop a solution that addresses the identified business challenges and supports the organization's strategic goals. The proposed system aims to improve operational efficiency, enhance service quality, reduce manual effort, and provide stakeholders with a reliable and user-friendly platform for managing business processes. The objectives defined in this section serve as measurable goals that guide project planning, development, and implementation.

4.1 Primary Business Objective

Objective ID	Business Objective	Expected Business Benefit
BO-01	Improve operational efficiency by automating manual business processes.	Reduced processing time and improved productivity.
BO-02	Provide a centralized system for managing business information and transactions.	Improved data consistency and accessibility.
BO-03	Enhance the overall user experience through a simple and intuitive interface.	Increased user satisfaction and system adoption.
BO-04	Improve data accuracy by minimizing manual data entry and processing errors.	Better decision-making based on reliable information.
BO-05	Strengthen security through controlled access and secure data management practices.	Protection of business and customer information.
BO-06	Enable real-time access to information and reporting.	Faster business decisions and improved operational visibility.

4.2 Strategic Objectives

The proposed solution is expected to contribute to the organization's long-term business strategy by supporting digital transformation initiatives, improving operational performance, and enabling scalable business growth. It also aims to provide a flexible foundation that can accommodate future business requirements without significant changes to the existing system.

4.3 Success Criteria

Criteria ID	Success Criterion	Measurement Method
SC-01	Business processes are completed with reduced manual intervention.	Process efficiency reports.
SC-02	Users can successfully perform their required business activities.	User Acceptance Testing (UAT).
SC-03	Information is available accurately and in a timely manner.	System reports and business validation.
SC-04	The solution supports the defined business requirements.	Requirement validation during project acceptance.
SC-05	Stakeholders approve the implemented solution.	Formal business sign-off.

4.4 Business Value

The implementation of the proposed solution is expected to deliver measurable business value by improving efficiency, enhancing data quality, supporting informed decision-making, reducing operational costs, and increasing stakeholder satisfaction. The system will provide a structured and standardized approach to managing business activities while supporting the organization's future growth and continuous improvement initiatives.

5. Project Scope

The Project Scope defines the boundaries of the Loan Payment Web System by identifying the features, functionalities, and business processes that will be included in the project, as well as those that are outside its scope. Clearly defining the scope helps establish stakeholder expectations, prevents scope creep, and ensures that the project team remains focused on delivering the agreed business objectives within the planned timeline and budget.

5.1 In Scope

Scope ID	Business Function	Description
IS-01	Identity Verification	Customer verifies using registered mobile number and card last 4 digits.
IS-02	CAPTCHA Validation	System validates reCAPTCHA before verification request.
IS-03	Loan Details Display	Customer can view total due, minimum due, due date, and payment status.
IS-04	Repayment Processing	Customer can make payment through simulated 1Pay flow.
IS-05	Payment Confirmation	System generates confirmation and receipt after successful payment.
IS-06	Session Security	JWT-authenticated flow and secure session-based access.
IS-07	Administrative Dashboard	Administrators can monitor loan repayments, customer accounts, and payment activities.
IS-08	Responsive Web Experience	Application usable on desktop and mobile web.

5.2 Out of Scope

Scope ID	Excluded Functionality	Reason
OS-01	Loan origination/application	Project scope is repayment only.
OS-02	Loan approval workflow	Managed in external/core banking systems.
OS-03	New user registration & forgot password	Not required in current assignment release.
OS-04	Admin dashboard	Deferred to future phase
OS-05	Enterprise reporting/MIS	Deferred to future phase
OS-06	Automated reminder notifications	Deferred due to external integration dependency
OS-07	Native mobile app	Current delivery is web-only

5.3 Scope Assumption

- Customers already possess an active loan account.
- Loan details are available within the organization's database.
- A secure payment gateway will be available for processing transactions.
- Users have valid authentication credentials.
- Internet connectivity is available during system usage.
- The project focuses on web-based access only.

5.4 Scope Constraints

Constraint	Description
Budget	The project must be completed within the approved budget.
Timeline	The solution must be delivered within the agreed project schedule.
Regulatory Compliance	The system must comply with applicable banking and data protection regulations.
Technology	The solution will be developed as a web application using the approved technology stack.
Third-Party Integration	Payment processing depends on the availability of the selected payment gateway.

5.5 Scope Diagram

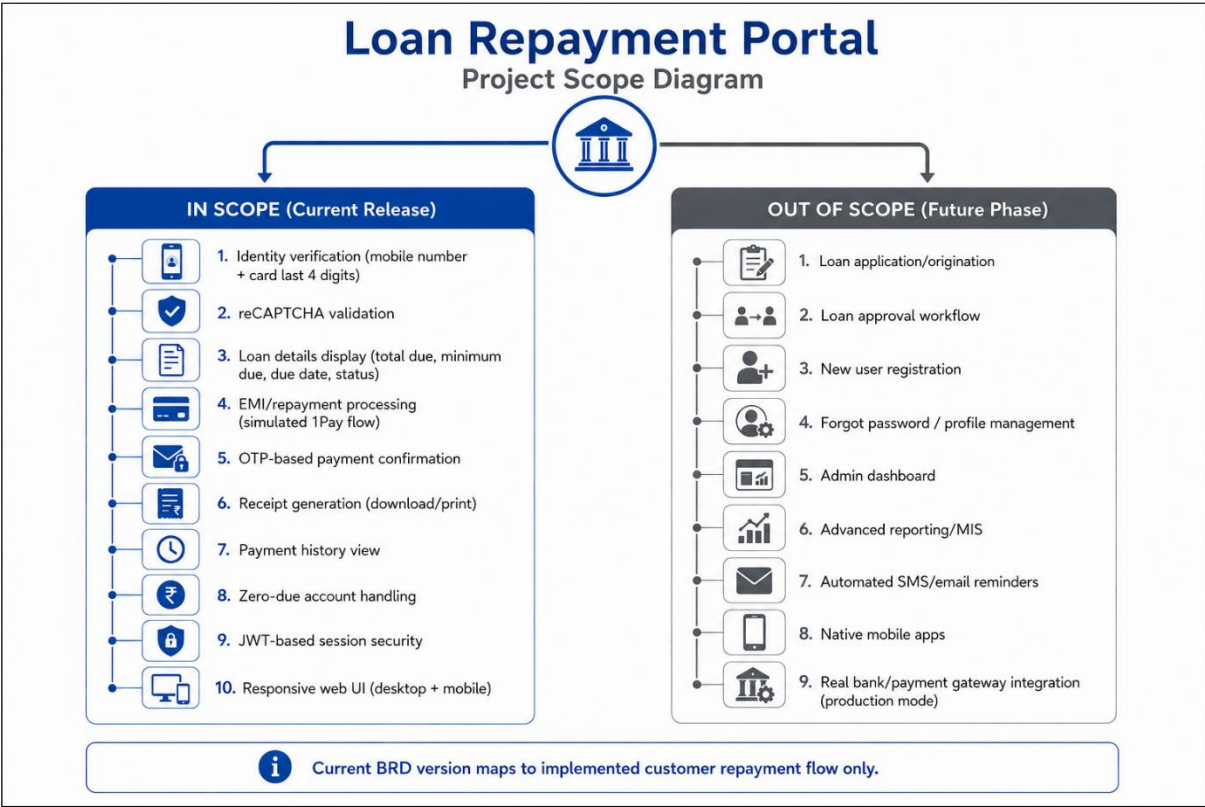


Figure 1. Project Scope

6. Stakeholder Analysis

Stakeholder analysis identifies the individuals, groups, and organizations that have an interest in or are affected by the Loan Payment Web System. Understanding stakeholder roles, responsibilities, expectations, and influence is essential for effective communication, requirement gathering, decision-making, and successful project delivery. This section defines the key stakeholders involved in the project and outlines their level of involvement throughout the project lifecycle.

6.1 Stakeholder Identification

Stakeholder	Role	Interest in the Project
Customer / Borrower	End User	Makes EMI payments, views loan information, and accesses payment history.
Bank Administrator	System Administrator	Manages customers, monitors payments, and generates reports.
Business Analyst	Requirement Owner	Gathers, analyzes, documents, and validates business requirements.
Project Manager	Project Management	Plans, monitors, and manages project execution.
Development Team	Implementation	Develops and implements the system based on approved requirements.
Quality Assurance Team	Testing	Validates that the solution meets business and functional requirements.
Finance Department	Business Operations	Verifies payment records and financial reconciliation.
Payment Gateway Provider	External Service	Processes secure online payment transactions.

6.2 Roles & Responsibilities

Stakeholder	Responsibilities
Customer	Register, view loan details, make EMI payments
Bank Administrator	Manage loans, customers, and reports
Business Analyst	Gather and document business requirements
Project Manager	Manage project scope, schedule, and risks
Development Team	Develop and maintain the application
QA Team	Test and validate the system
Finance Department	Verify payments and financial records

6.3 Stakeholder Communication

Stakeholder	Communication Method	Frequency
Business Sponsor	Review Meeting	Monthly
Project Manager	Status Meeting	Weekly
Business Analyst	Requirement Workshop	As Required
Development Team	Team Meeting	Daily
QA Team	Test Review	Weekly

6.4 Stakeholder Matrix

Stakeholder	Influence	Interest
Business Sponsor	High	High
Project Manager	High	High
Business Analyst	High	High
Customer	Medium	High
Development Team	High	Medium
QA Team	Medium	Medium

7. Business Requirements

Business Requirements define the high-level business needs that the Loan Payment Web System must fulfil to achieve the project's objectives. These requirements focus on business outcomes rather than technical implementation.

7.1 Business Requirement Summary

BR ID	Business Requirement	Priority
BR-01	Verify customer identity using mobile number and card last 4 digits.	High
BR-02	Validate reCAPTCHA before verification.	High
BR-03	Show loan due details and payment status.	High
BR-04	Enable customer repayment through online payment flow.	High
BR-05	Generate receipt after successful payment.	High
BR-06	Maintain and show payment history.	High
BR-07	Handle zero-due account scenario.	Medium
BR-08	Protect access using secure authenticated sessions	High

7.2 Requirement Priority

Priority	Description
High	Mandatory for the initial release.
Medium	Important but can be included after core features.
Low	Future enhancement.

7.3 Business Requirement Details

BR ID	Description	Business Value
BR-01	Customer shall access account using valid registered credentials.	Prevent unauthorized access.
BR-02	System shall verify human interaction using reCAPTCHA.	Reduce automated misuse.
BR-03	System shall display current dues and repayment information.	Improve transparency.
BR-04	Customer shall be able to make repayment online.	Improve convenience and efficiency.
BR-05	System shall generate receipt for successful payment.	Provide payment proof.
BR-06	System shall maintain complete payment transaction history.	Enable auditability and support.
BR-07	System shall show no-payment-required state for zero dues.	Improve user experience.
BR-08	System shall use secure token/session controls for protected endpoints.	Protect sensitive flows.

8. Functional Requirements

Functional Requirements describe the features and functionalities that the Loan Payment Web System must provide to meet the defined business requirements.

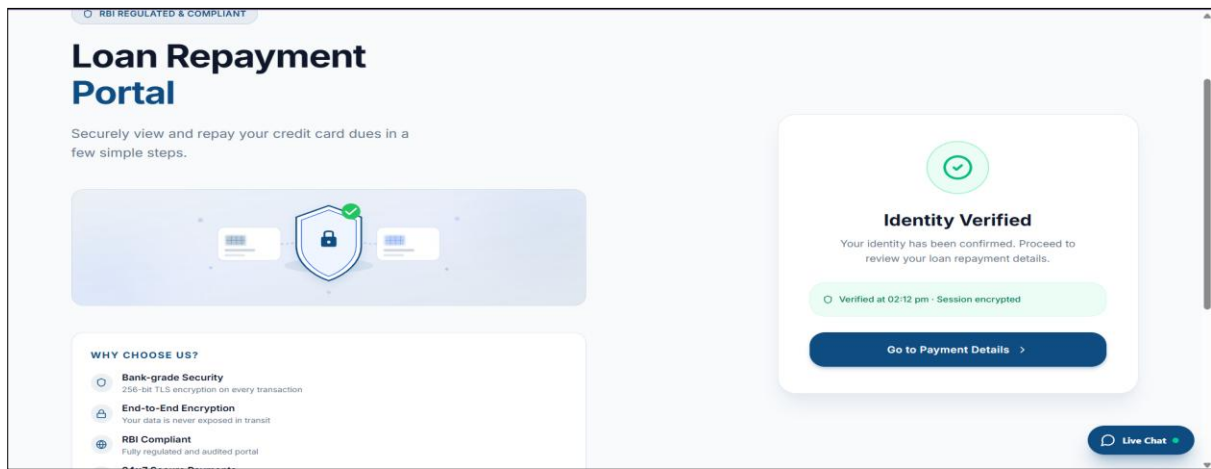


Figure 2. Verification Page

8.1 Functional Requirement Summary

FR ID	Module	Functional Requirement	Priority
FR-01	Verification	Verify customer with mobile + card last4.	High
FR-02	Verification	Validate reCAPTCHA token.	High
FR-03	Loan Details	Display due amount, minimum due, and due date.	High
FR-04	Payment	Allow user to select repayment amount option.	High
FR-05	Payment Gateway	Process simulated 1Pay payment and OTP flow.	High
FR-06	Receipt	Generate/display receipt after success.	High
FR-07	History	Show payment history list and status.	High
FR-08	Session	Maintain and validate secure session tokens.	High
FR-09	Error Handling	Show payment failure with retry path.	Medium
FR-10	Zero Due	Prevent payment when outstanding is zero.	Medium

8.2 Module-wise Functional Requirements

Verification Management

- Identity verification via registered mobile and card last4
- CAPTCHA validation before verification submits

Loan Management

- View loan due details
- View due date and account status

Payment Management

- Select repayment amount (minimum/full/custom)
- Simulated payment flow with OTP verification
- Success/failure path handling

Receipt & History

- Receipt generation and display
- Payment history with transaction records

Security & Session

- Token-protected API access
- Session expiry/refresh handling

8.3 Functional Requirement Mapping

Business Requirement	Related Functional Requirement(s)
BR-01	FR-01, FR-02
BR-02	FR-04, FR-05
BR-03	FR-06
BR-04	FR-08
BR-05	FR-07
BR-06	FR-09
BR-07	FR-10
BR-08	FR-11

9. Non-Functional Requirements

Non-Functional Requirements define the quality standards and performance expectations of the Loan Payment Web System, including security, performance, reliability, usability, and availability.

9.1 Non-Functional Requirement Summary

NFR ID	Category	Requirement
NFR-01	Performance	Core pages (verification, loan details, payment pages) should load within 3 seconds under normal network conditions.
NFR-02	Performance	Verification and payment API responses should return within 5 seconds under normal load.
NFR-03	Security	Users must be verified before accessing loan/payment features.
NFR-04	Security	Sensitive data must be encrypted in transit and protected in processing/storage flows.
NFR-05	Availability	Target service availability is 99.5% monthly uptime (subject to hosting tier limits).
NFR-06	Reliability	Payment records must be stored accurately and remain query able in payment history.
NFR-07	Usability	Application must provide intuitive, low-friction repayment flow for non-technical users.
NFR-08	Compatibility	Support latest versions of major browsers (Chrome, Edge, Firefox, Safari).
NFR-09	Responsiveness	UI must be usable across desktop and mobile screen sizes.
NFR-10	Compliance	System must follow applicable security and audit expectations for financial transaction systems.

9.2 Requirement Categories

Category	Description
Performance	Fast response time and efficient transaction processing.
Security	Secure authentication, authorization, and data protection.
Availability	High system uptime with minimal downtime.
Reliability	Accurate processing and regular data backup.
Usability	Easy-to-use interface for customers and administrators.
Compatibility	Support for major web browsers and responsive design.
Scalability	Ability to accommodate increased users and transactions.
Compliance	Adherence to organizational and regulatory standards.

9.3 Priority Classification

Priority	Description
High	Mandatory for system operation and security.
Medium	Important for performance and user experience.
Low	Future enhancement or optimization.

10. Business Rules

Business Rules define the policies and constraints that govern the business processes of the Loan Payment Web System. These rules ensure consistent, secure, and accurate processing of loan payments.

10.1 Business Rule Summary

Rule ID	Business Rule	Category
BRL-01	Only verified users can access loan and payment steps.	Security
BRL-02	Payments are allowed only for active customer loan records.	Loan Management
BRL-03	Payment amount cannot exceed outstanding due amount.	Payment
BRL-04	Minimum due rules apply when “Minimum Due” option is selected.	Payment
BRL-05	Loan/payment records update only after successful payment confirmation.	Loan Management
BRL-06	Receipt must be generated for every successful transaction.	Payment
BRL-07	Each payment attempt (success/failure) must be logged for audit trail.	Compliance
BRL-08	If outstanding due is zero, payment initiation must be blocked.	Loan Management
BRL-09	Session token is mandatory for secured API operations after verification.	Security
BRL-10	CAPTCHA validation is required before verification (unless approved demo mode is enabled).	Security

10.2 Business Rule Categories

Category	Purpose
Security	Protect user accounts and business data.
Loan Management	Ensure accurate loan processing and updates.
Payment	Validate and process loan payments correctly.
Administration	Control administrative access and operations.
Compliance	Maintain audit records and regulatory compliance.
Notification	Keep customers informed about payment activities.

10.3 Rule Priority

Priority	Description
High	Mandatory business rule required for system operation.
Medium	Important rule supporting business processes.
Low	Optional rule for future enhancement.

11. Assumptions and Constraints

This section outlines the assumptions made during project planning and the constraints that may impact the successful delivery of the Loan Payment Web System.

11.1 Assumptions

ID	Assumption
A-01	Customer records and loan data are already available in the database.
A-02	Users have valid registered mobile number and card last4 details.
A-03	Internet access is available during verification and payment.
A-04	Third-party CAPTCHA and payment simulation services are reachable.
A-05	Users access the system via modern web browsers.
A-06	Current release focuses on customer repayment workflow only

11.2 Constraints

ID	Constraint
C-01	Delivery timeline and assignment scope are fixed.
C-02	Current hosting runs on free-tier cloud plans with cold-start limitations.
C-03	Real banking gateway integration is outside current phase.
C-04	Notification automation (SMS/email reminders) is deferred.
C-05	Any additional modules (admin/reporting suite) require separate phase planning.

11.3 Project Dependencies

Dependency	Description
Supabase Database	Required for customer, loan, and transaction data.
Backend API (Render)	Required for verification, loan details, payment, history APIs.
Frontend App (Vercel)	Required for customer interaction and workflow UI.
reCAPTCHA Service	Required for human-verification in production mode.
Payment Simulation Logic	Required to execute demo OTP-based payment flow.

12. Risks

This section identifies potential risks that may impact the successful implementation of the Loan Payment Web System and outlines mitigation strategies to minimize their impact.

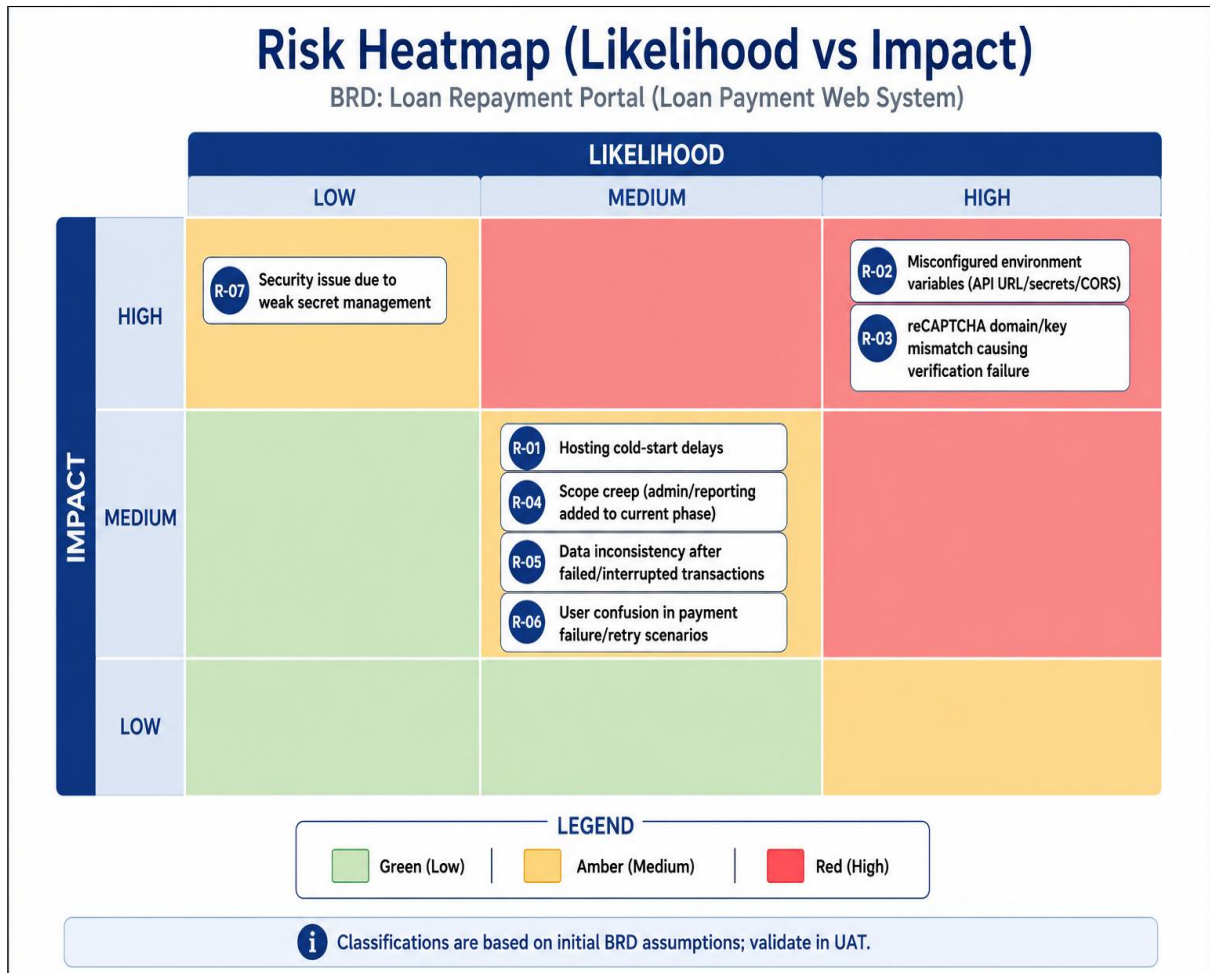


Figure 3. Risk Heatmap

12.1 Risk Register

Risk ID	Risk Description	Impact	Mitigation Strategy	Owner
R-01	Hosting cold-start delays on free plan	Medium	Display clear loading states; consider paid plan for production.	Project Manager
R-02	Misconfigured environment variables (API URL, secrets, CORS)	High	Deployment checklist + post-deploy smoke testing.	Dev Team
R-03	reCAPTCHA domain/key mismatch causing verification failure	High	Maintain correct domain whitelist and key mapping.	Dev Team
R-04	Scope creep from adding admin/reporting during current phase	Medium	Freeze current release scope and track future-phase backlog.	BA / PM
R-05	Data inconsistency after failed/interrupted transactions	Medium	Ensure transactional updates + retry-safe APIs.	Dev Team
R-06	User confusion in payment failure/retry scenarios	Medium	Improve error messages and guided retry path.	UX / QA
R-07	Security issue due to weak secret management	High	Use secure env management and periodic secret rotation	Security Owner

12.2 Risk Priority

Priority	Description
High	Immediate attention required; may significantly impact the project.
Medium	Moderate impact; requires monitoring and mitigation.
Low	Minor impact; monitor throughout the project lifecycle.

13. Use Case Summary

This section provides a high-level overview of the primary use cases supported by the Loan Payment Web System. These use cases represent the key interactions between users and the system.

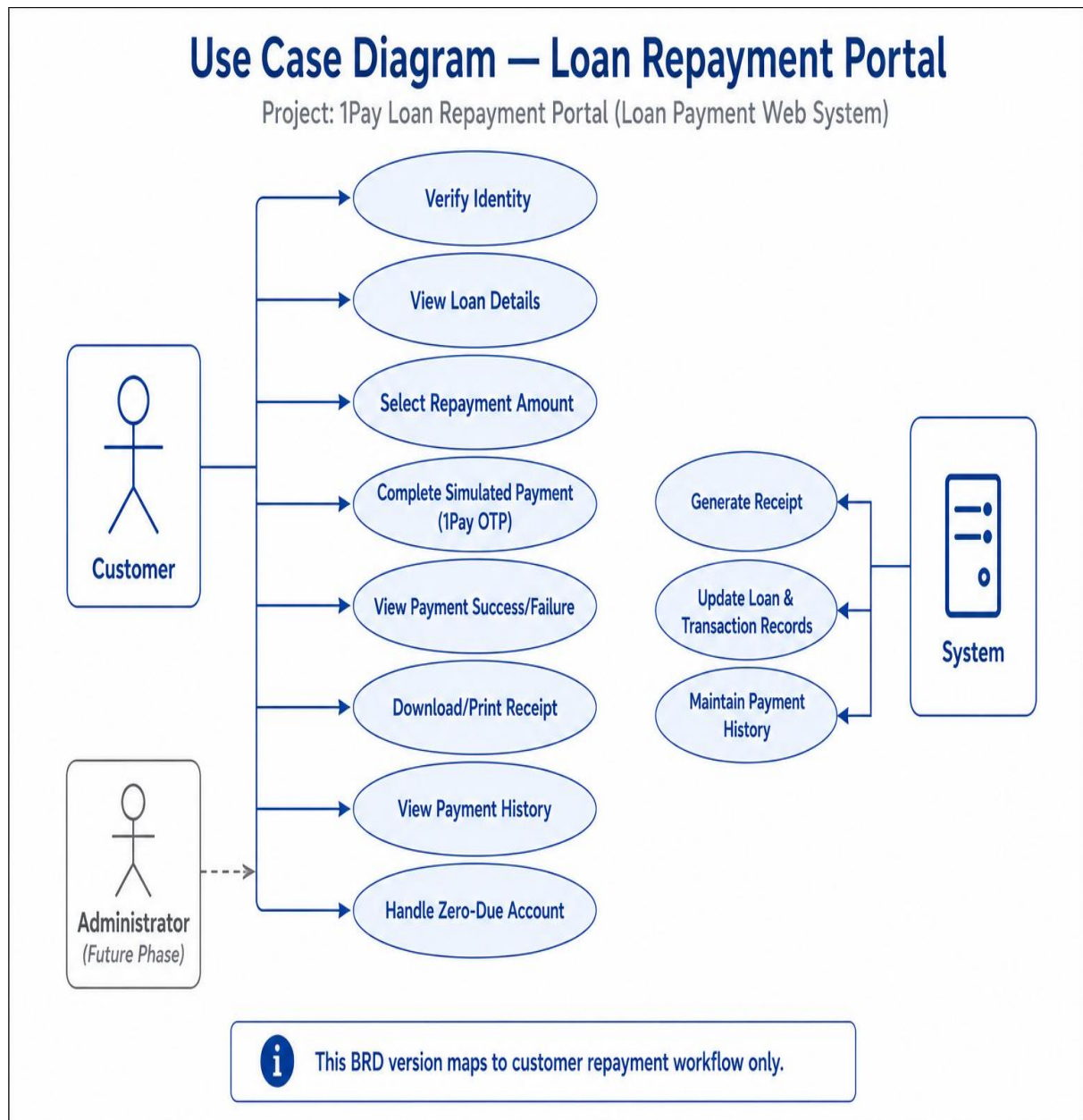


Figure 4. Use Case Diagram

13.1 Use Case Summary

Use Case ID	Use Case Name	Primary Actor	Description
UC-01	Verify Identity	Customer	User enters mobile + card last4 and passes CAPTCHA.
UC-02	View Loan Details	Customer	System shows due details after successful verification.
UC-03	Select Repayment Amount	Customer	User selects minimum/full/custom amount.
UC-04	Complete Simulated Payment	Customer	User performs OTP-based payment in 1Pay simulation.
UC-05	View Payment Outcome	Customer	User sees success/failure with next-action options.
UC-06	Download/Print Receipt.	Customer	User accesses generated receipt after success.
UC-07	View Payment History	Customer	User checks previous transaction records.
UC-08	Handle Zero Due	Customer	System shows no-payment-required state.

13.2 Actor Summary

Actor	Responsibilities
Customer	Verify identity, view dues, make payment, view receipt/history.
Payment Simulation Service	Process OTP-based demo transaction flow.
Backend API	Validate verification, enforce business rules, maintain records.
Support/Operations (Business Role)	Review logs/history for issue resolution (outside direct app UI in this release).

14. Process Flow

The process flow illustrates the high-level business process for loan repayment, from user authentication to successful payment completion and record update.

14.1 Loan Payment Process Flow

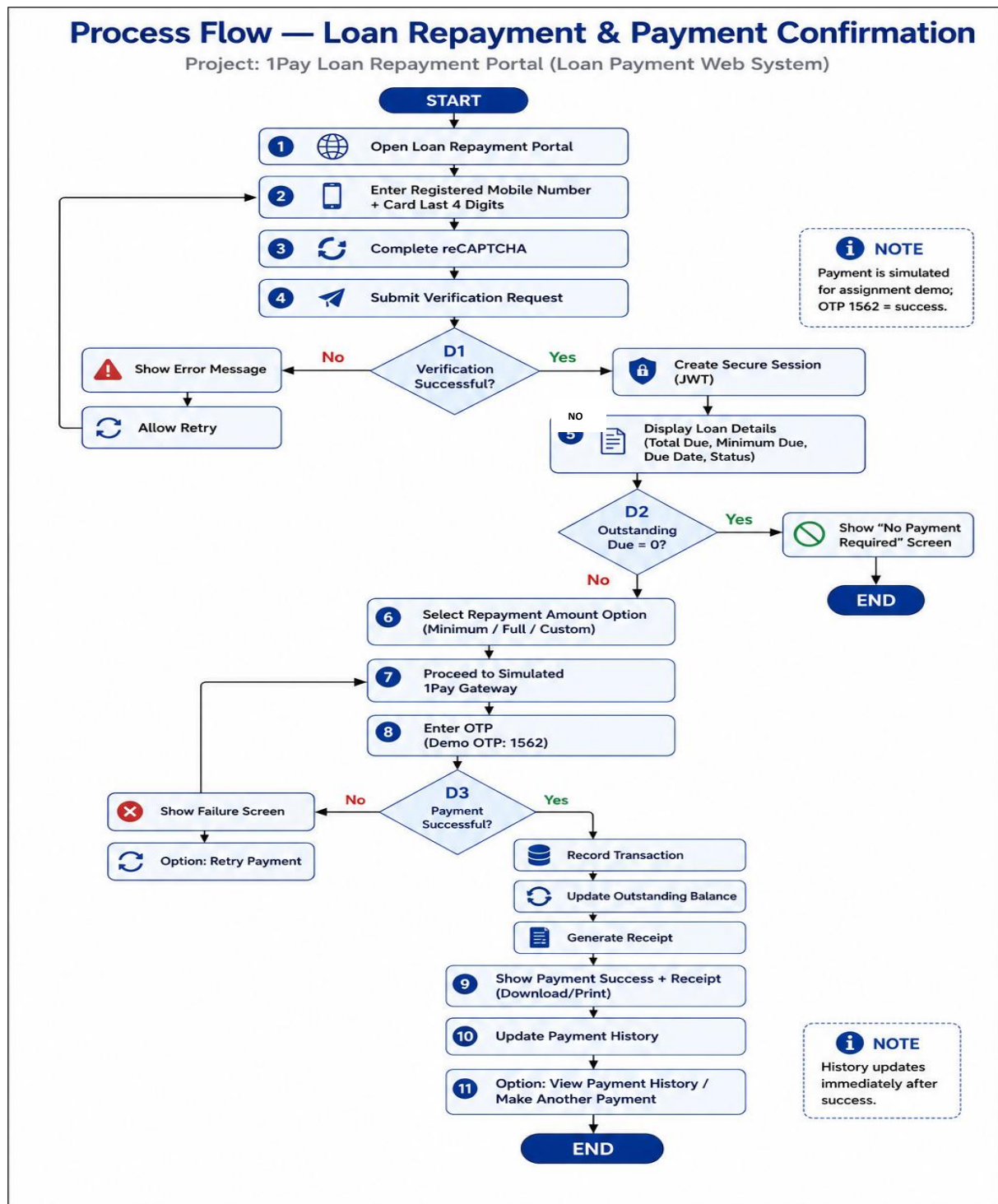


Figure 5. Process Flow

14.2 Process Description

Steps	Description
1	Customer opens the repayment portal.
2	Customer enters registered mobile number and card last4.
3	Customer completes CAPTCHA and submits verification.
4	System validates credentials and creates authenticated session.
5	System displays loan details and available repayment options.
6	Customer selects amount and proceeds to payment gateway simulation.
7	Customer enters OTP and confirms payment.
8	System records transaction outcome (success/failure).
9	On success, receipt is generated and payment history is updated.
10	User may view history or start another payment; if zero due, no-payment-required state is shown.

15. Reporting Requirements

This section identifies the key reports required by stakeholders to monitor loan payments, customer activity, and overall system performance.

15.1 Reporting Requirement Summary

Report ID	Report Name	Purpose	User	Frequency
RR-01	Customer Payment History	Enables customers to view previous repayment transactions and statuses.	Customer	On Demand
RR-02	Payment Receipt	Provides proof of successful payment with transaction reference details.	Customer	Per Successful Payment
RR-03	Transaction Outcome View	Displays latest transaction result (success or failure) for customer confirmation and support reference.	Customer / Support	On Demand

15.2 Report Distribution

Stakeholder	Report
Customer	Customer Payment History, Payment Receipt.
Support / Loan Operations	Transaction Outcome View.

Note: Organisation-level reporting such as collection analysis, overdue dashboards, branch performance reports, and management MIS are not included in the current release and are planned for a future phase.

16. Acceptance Criteria

Acceptance Criteria define the conditions that must be met before the Loan Payment Web System is accepted by the business stakeholders and approved for deployment.

16.1 Acceptance Criteria Summary

AC ID	Acceptance Criteria	Owner
AC-001	Customer identity verification succeeds with valid registered details and fails with invalid details.	Business Sponsor
AC-002	reCAPTCHA validation is enforced before verification submission, or approved demo fallback mode is available for assignment testing.	Information Security Team
AC-003	System displays correct loan details including total due, minimum due, due date, and status after successful verification.	Loan Operations
AC-004	Customer can successfully complete the simulated payment flow using OTP-based confirmation.	Finance Department
AC-005	Successful payment generates a downloadable or printable payment receipt.	Business Sponsor
AC-006	Payment history is updated immediately after successful transaction and shows correct transaction status.	Loan Operations
AC-007	Zero-due customer accounts display a “No Payment Required” state and block payment initiation.	Operations Manager
AC-008	Core security controls such as authenticated APIs, secure sessions, and protected payment flow are verified during testing.	Information Security Team
AC-009	End-to-end user acceptance testing is completed successfully for all in-scope customer repayment features.	Business Sponsor

16.2 Project Acceptance Conditions

Condition	Status
Business requirements implemented for current in-scope release	Required
Functional requirements completed for current release	Required
Non-functional requirements validated at basic deployment/UAT level	Required
User Acceptance Testing completed	Required
Stakeholder approval received	Required
Frontend and backend deployment completed successfully	Required

17. Glossary

This section defines the key business and technical terms used throughout this Business Requirements Document to ensure a common understanding among all stakeholders.

17.1 Glossary of Terms

Term	Definition
1Pay	Simulation Demo payment flow used for assignment/testing purposes instead of real payment gateway integration
OTP	One-Time Password used to confirm payment in the simulated gateway
No Payment Required	A system state displayed when the customer has zero outstanding balance
Payment History	List of previous repayment transactions and their outcomes
Receipt	Downloadable/printable proof generated after successful payment
reCAPTCHA	Human verification mechanism used to reduce bot or automated misuse
Session Token	Secure token generated after successful identity verification to access protected features

18. Appendix A. Deployment & Demo References

GitHub Repository

- <https://github.com/Goutamp241/loan-payment-portal>

Frontend Deployment (Vercel)

- <https://loan-payment-portal-sand.vercel.app>

Backend Deployment (Render)

- <https://loan-payment-portal.onrender.com>

Demo Credentials

- Customer 1: Mobile 9876543210, Card Last 4 4821
- Zero-Due Customer: Mobile 9660683381, Card Last 4 1567

Simulated Payment OTP

OTP: 1562